ACICP302ICS – Module 1

Practical Lab Report

Analyze, Test & Defend the Supplied Modbus/TCP OT Simulation

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Practical Lab Assessment
Modbus/TCP Operational Technology Security

Item	Recorded Value
Lab VM	ACICP302ICS-Ubuntu-OT-Lab
Operating System	Ubuntu 24.04.5 LTS Desktop
Host name	nando-VirtualBox
Scope	Local loopback only
Allowed target	127.0.0.1
Course source	`labs/ot-security/student-lab-source/`
Source commit	`2a66f9bcf253eb3fe31c68ad52794cd535421da8`
Network during lab	Disconnected
Demonstrations	False-data, slow-drift, replay

1 Executive Summary

This practical assessed a supplied Modbus/TCP OT simulation using a controlled Ubuntu VM. The work was performed only against loopback services on `127.0.0.1`; the archived source was not executed.

The simulation was analysed in both direct and guarded paths. Normal Modbus/TCP traffic was captured and correlated with the supplied register map. Three bounded demonstrations were then performed using the supplied scripts: false-data injection, slow drift, and replay.

The guard successfully blocked the unsafe false-data and slow-drift values. The replay demonstration did not produce a corresponding guard block in the observed event log, identifying a limitation of the supplied rule set.

After testing, the dashboards returned to normal operating ranges, all four services were stopped, the expected ports were confirmed closed, and the VM network remained isolated.

2 Activity 00 – Preflight and Safety

The lab was executed in the assigned Ubuntu VM using the canonical student source.

Check	Result
Correct VM	Confirmed
Canonical source	Confirmed
Source commit	`2a66f9bcf253eb3fe31c68ad52794cd535421da8`
Archived source executed	No
Service/client target	`127.0.0.1`
Network adapter during simulation	Disconnected
Bridged/external target	Not used
Port forwarding	Not used
Clean baseline snapshot	Created

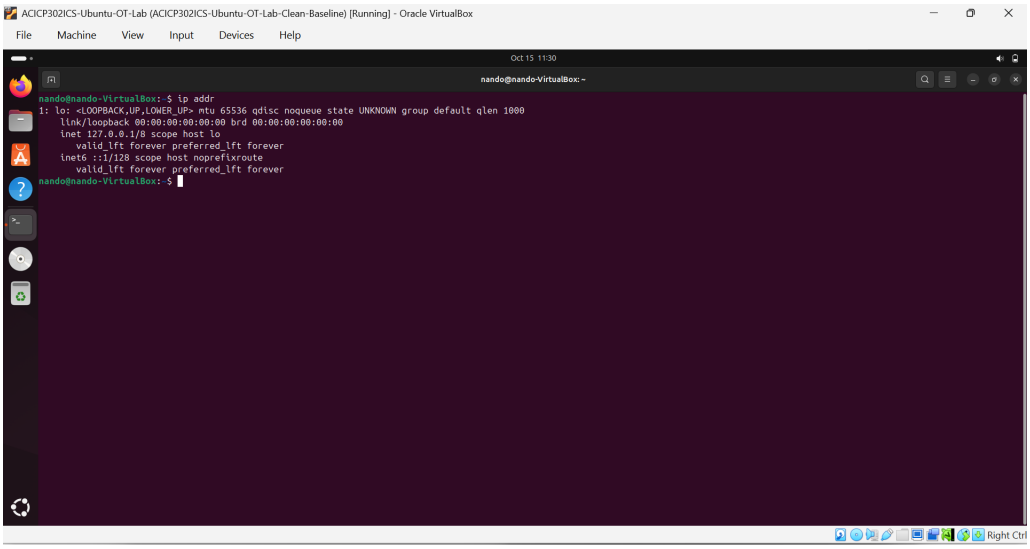


Figure 1: VM network isolation and loopback-only configuration.

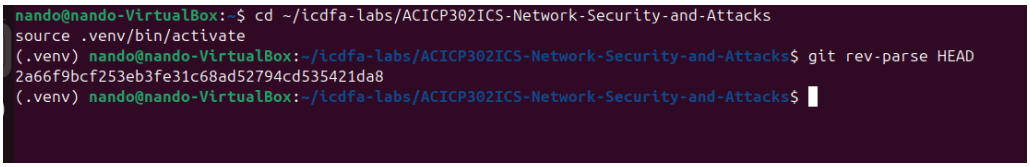


Figure 2: Recorded course source commit.

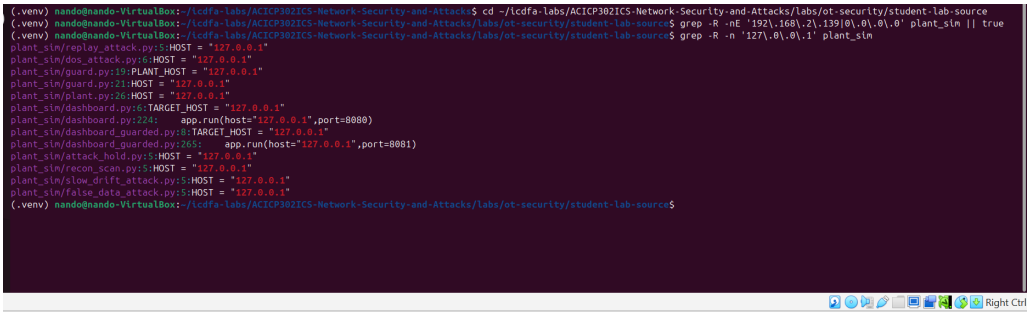


Figure 3: Safety check of the canonical student source.

Safety boundary: All testing remained inside the assigned VM and targeted only the supplied

loopback simulation.

3 Activity 01 – Architecture and Plant Baseline

The supplied simulation consists of a simulated plant, a Modbus/TCP service, a direct dashboard, a rule-based guard, and a guarded dashboard.

Component	Endpoint	Role
Plant	127.0.0.1:5020	Modbus/TCP process simulation
Guard	127.0.0.1:5021	Validates writes before forwarding
Direct dashboard	127.0.0.1:8080	Displays plant state without guard
Guarded dashboard	127.0.0.1:8081	Displays guarded process state

Data flow:

$$Plant \rightarrow Modbus/TCP \rightarrow \{\ Directdashboard Guard \rightarrow Guardeddashboard$$

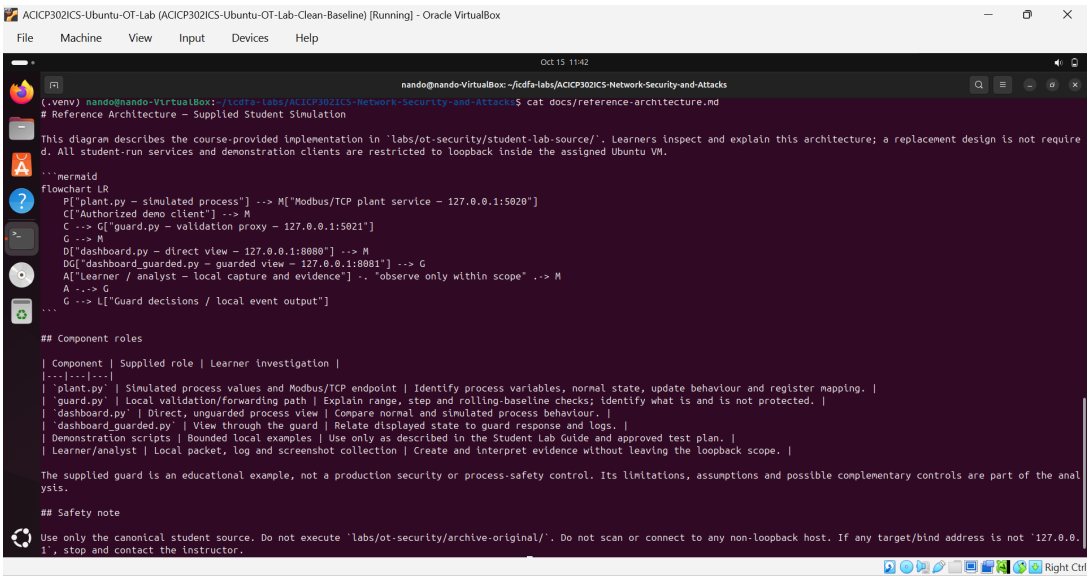


Figure 4: Supplied OT simulation architecture and data flow.

Normal observations were approximately within the supplied safe operating ranges:

Variable	Normal range	Observed baseline
Temperature	60–75°C	Approximately 67–74°C
Pressure	30–40 bar	Approximately 34–39 bar
Flow	50–150 Nm^3/h	Approximately 90–143 Nm^3/h

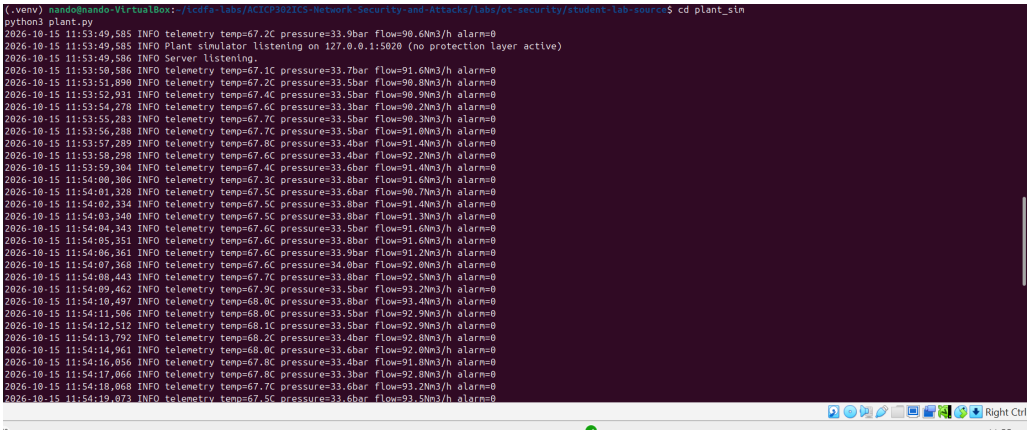


Figure 5: Normal plant baseline.

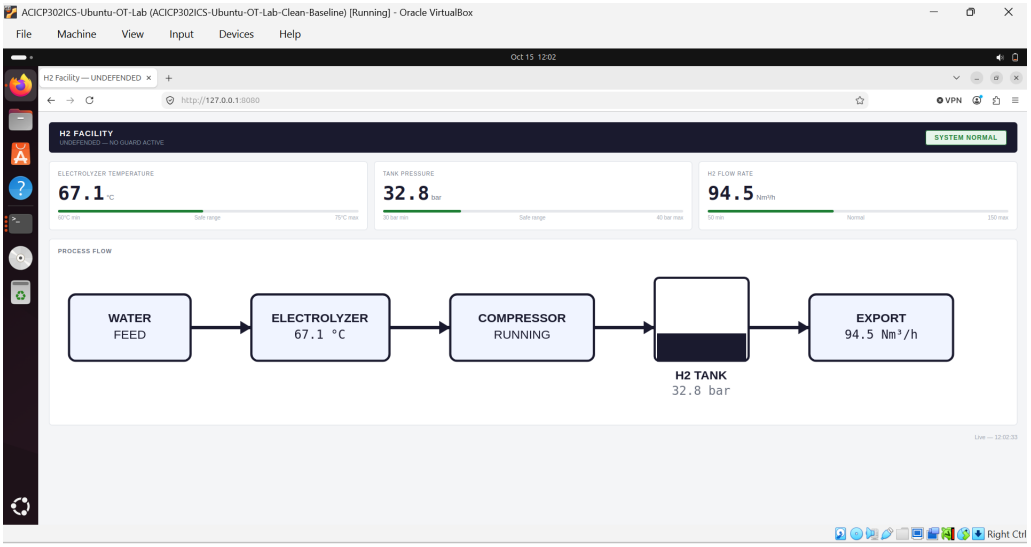


Figure 6: Direct dashboard during normal operation.

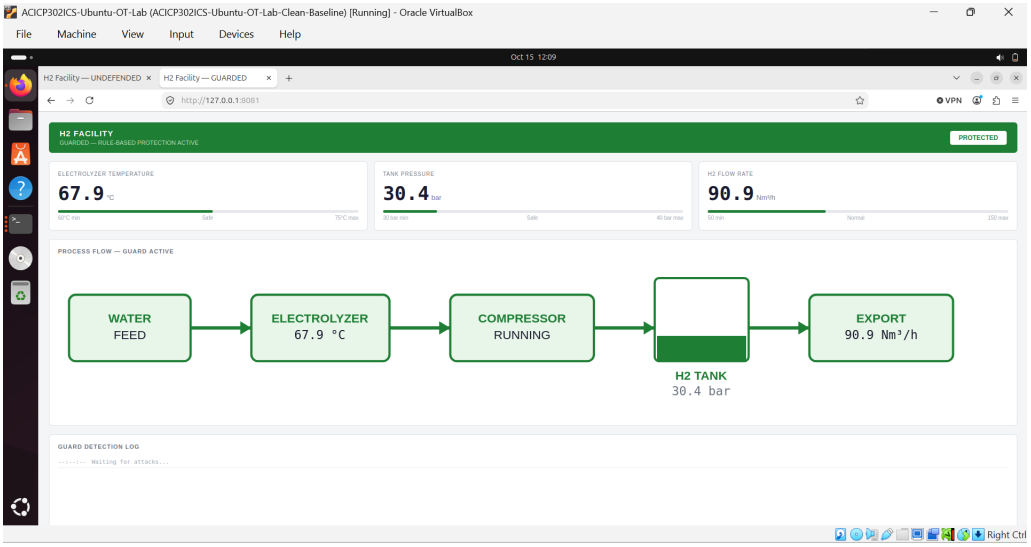


Figure 7: Guarded dashboard during normal operation.

4 Activity 02 – Modbus/TCP Data Model

The supplied plant uses four holding registers.

Register	Process value	Display scale	Safe range
0	Temperature	Register/10	60–75°C
1	Pressure	Register/10	30–40 bar
2	Flow	Register/10	50–150 Nm^3/h
3	Alarm	Integer flag	Source-defined

The plant updates its process values periodically. Temperature, pressure and flow are represented as integer register values and displayed after division by 10.

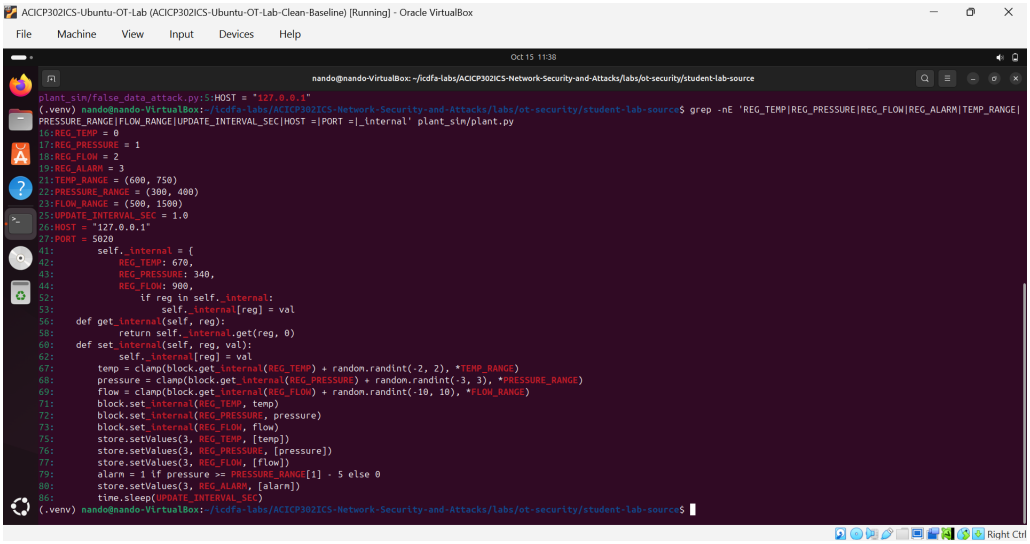


Figure 8: Register and process data model extracted from the supplied source.

5 Activity 03 – Normal Modbus/TCP Packet Analysis

Normal traffic was captured on the guest loopback interface `lo` using the documented Modbus/TCP port.

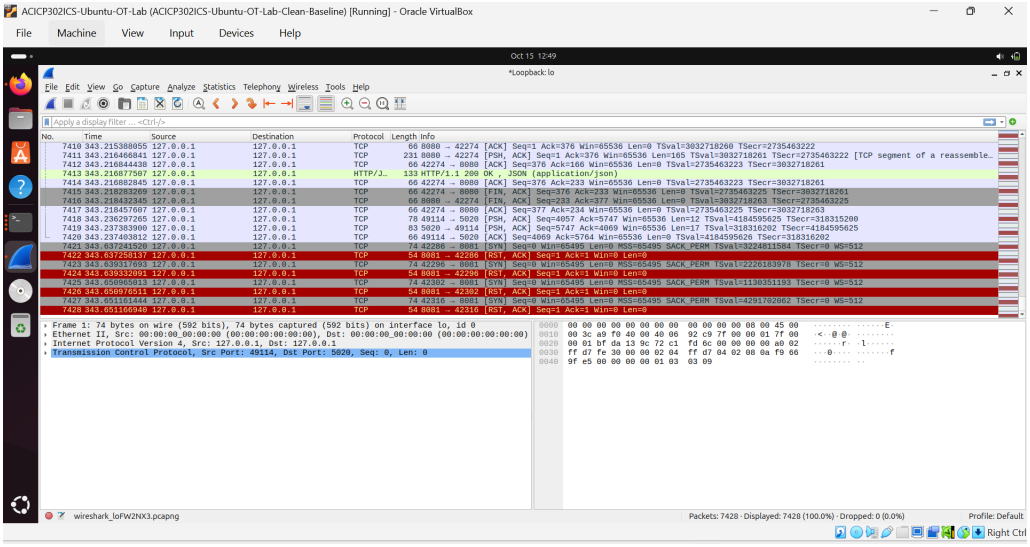


Figure 9: Normal Modbus/TCP packet capture on the loopback interface.

One observed request was:

```
00 01 00 00 00 06 01 03 00 00 00 04
```

Interpretation:

Field	Interpretation
Transaction ID	1
Protocol ID	0 (Modbus/TCP)
Length	6
Unit ID	1
Function code	03 – Read Holding Registers
Starting register	0
Quantity	4 registers

The corresponding response contained eight data bytes, representing four 16-bit register values.

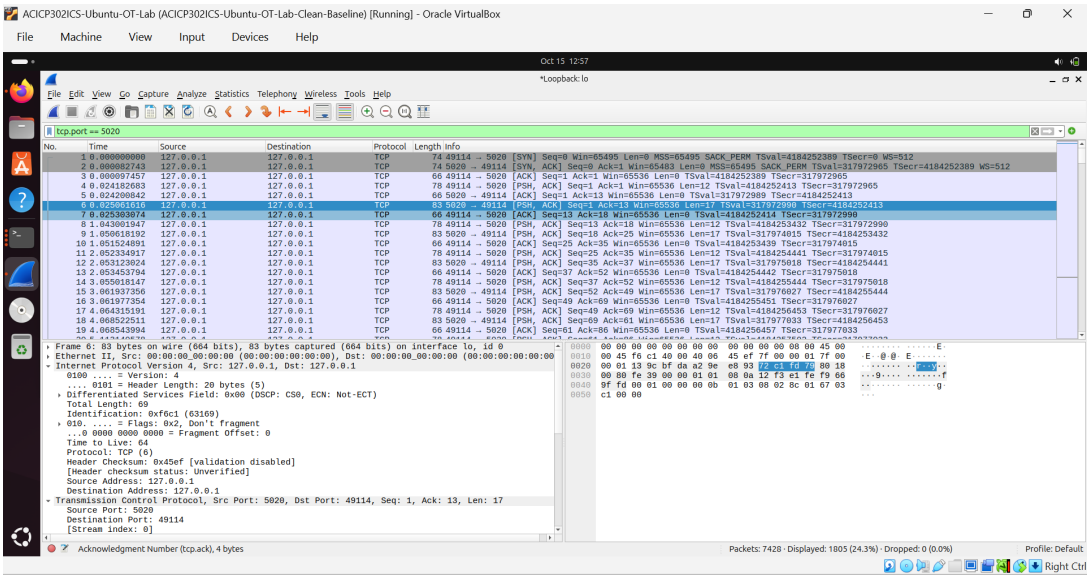


Figure 10: Matched Modbus/TCP request and response.

What the capture proves: A normal Modbus/TCP client successfully requested registers 0–3 from the local plant and received the corresponding register data.

What it does not prove: The capture does not by itself prove authentication, authorization, or protection against malicious writes.

6 Activity 04 – Local Exposure Mapping

The local service inventory identified the following expected lab endpoints.

Endpoint	Protocol	Role	Security relevance
127.0.0.1:5020	Modbus/TCP	Plant	Direct process interaction
127.0.0.1:5021	Modbus/TCP	Guard	Controlled process interaction
127.0.0.1:8080	HTTP	Direct UI	Unprotected process view/path
127.0.0.1:8081	HTTP	Guarded UI	Protected simulation view

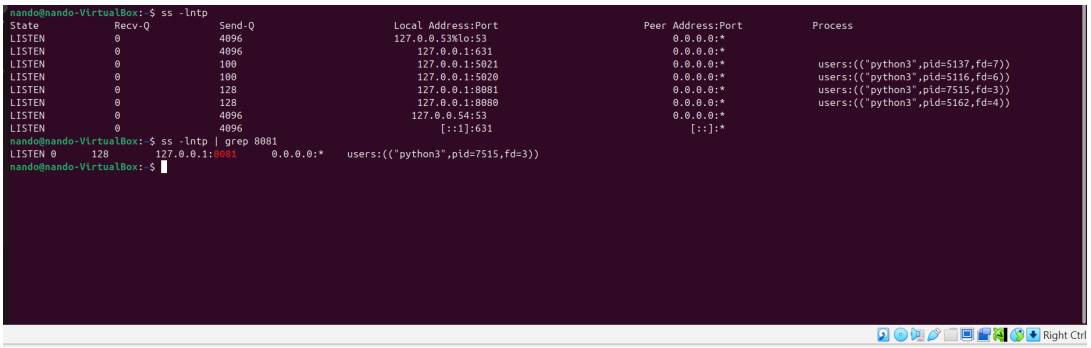


Figure 11: Local loopback service exposure mapping.

The main risk identified is unauthorized or unsafe Modbus writes if a client can reach the direct plant path. The guard reduces this risk but does not provide complete security controls.

7 Activity 05 – Analysis of the Supplied Guard

The supplied guard validates process writes before forwarding them to the plant.

Check	Input	Condition	Action
Range	New value	Outside safe range	Block
Step	Current/new value	Step exceeds limit	Block
Drift	Recent accepted values	Rolling drift exceeds limit	Block
Accepted write	Valid value	All checks pass	Forward to plant

Key limits observed in the supplied implementation include:

Variable	Maximum step	Drift limit
Temperature	3°C	2°C
Pressure	5 bar	2 bar
Flow	10 Nm^3/h	4 Nm^3/h

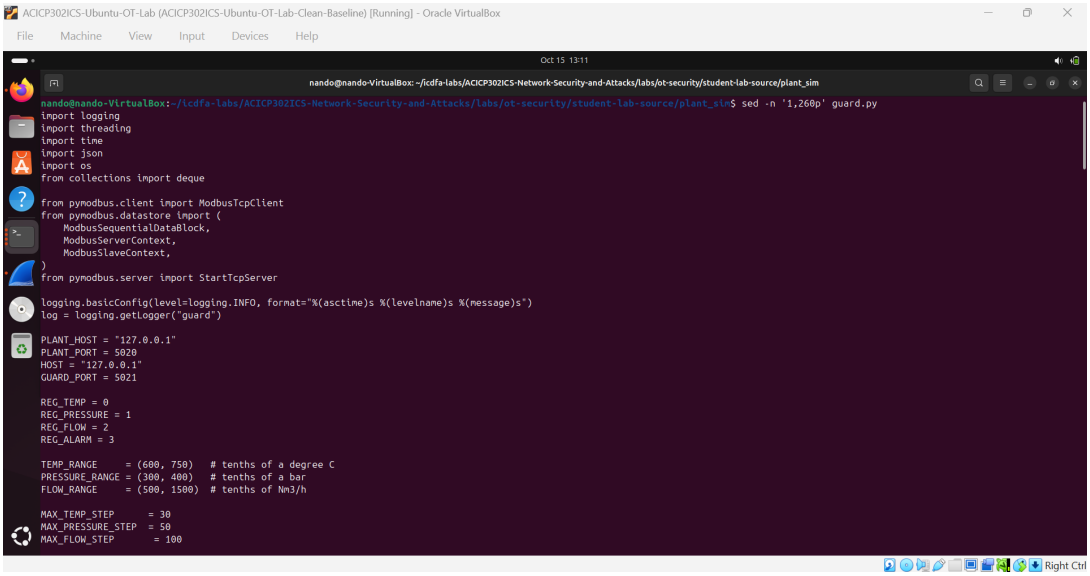


Figure 12: Guard checks identified from the supplied source.

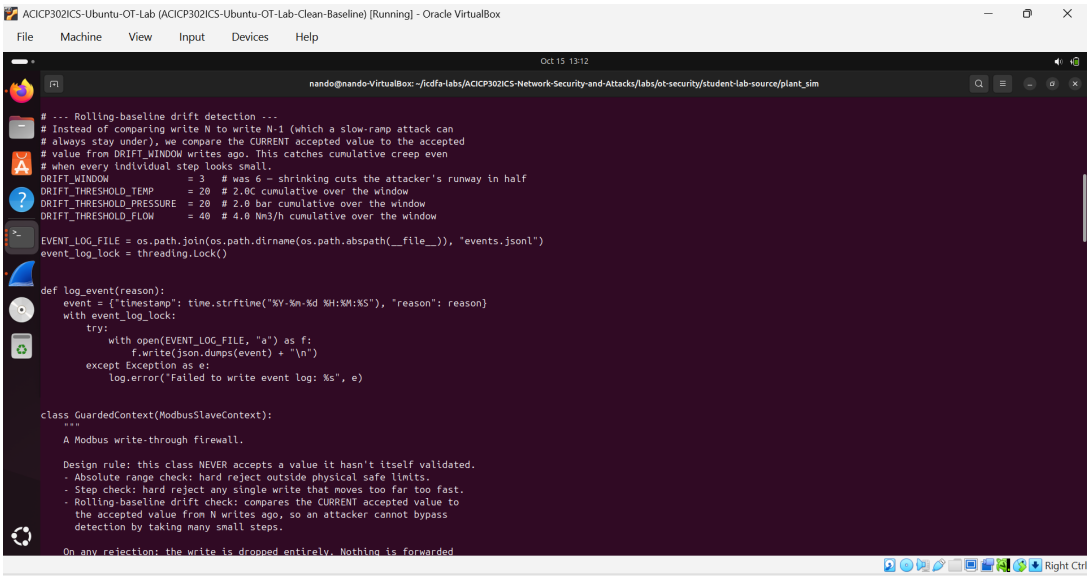


Figure 13: Guard limitations identified during source analysis.

Limitations:

- No demonstrated client authentication or authorization mechanism.
- The direct Modbus path remains available inside the simulation.
- Rule thresholds are fixed and require appropriate tuning.
- The demonstrated rules focus on individual process values.
- Replay/duplicate-command detection is not explicitly implemented.
- The guard is an educational simulation and not a production process safety mechanism.

8 Activity 06 – Controlled Security Validation

Three supplied demonstrations were executed one at a time using the supplied parameters.

8.1 Test Plan

Scenario	Target	Expected	Stop/Recovery
False data	5020/5021	Direct path reaches unsafe values; guard blocks	Stop if unstable; verify recovery
Slow drift	5020/5021	Direct path changes pressure; guard blocks unsafe progression	Stop if unstable; verify recovery
Replay	5020/5021	Repeated write is attempted; observe whether guard detects it	Verify final process state

8.2 Test 1 – False-Data Attack

The supplied script attempted:

- Temperature: 110.0°C
- Pressure: 55.0 bar
- Flow: 200.0 Nm^3/h

Direct path result: Unsafe values reached the plant.

Guarded path result: All three unsafe values were blocked.

The observed guard events reported range, step and cumulative-drift violations.

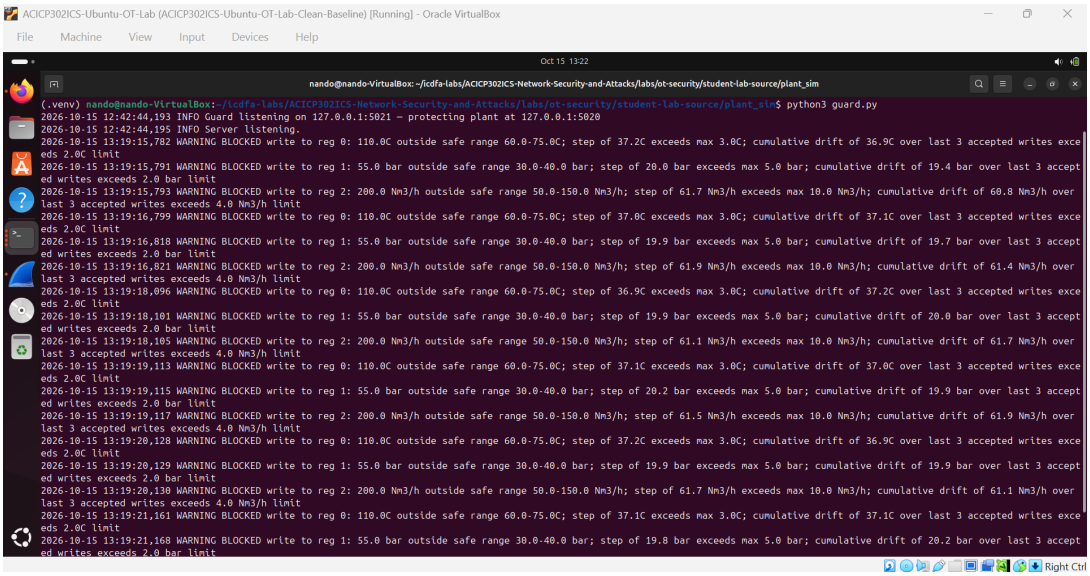


Figure 14: False-data test – guard blocking evidence.

8.3 Test 2 – Slow-Drift Attack

The supplied script increased pressure gradually from approximately 34.7 bar and attempted values up to approximately 46.7 bar on the direct path.

On the guarded path, attempted values from 40.4 to 51.4 bar were blocked.

The guard initially rejected values because they were outside the safe range. As the attempted drift increased, cumulative-drift and step-size checks also triggered.

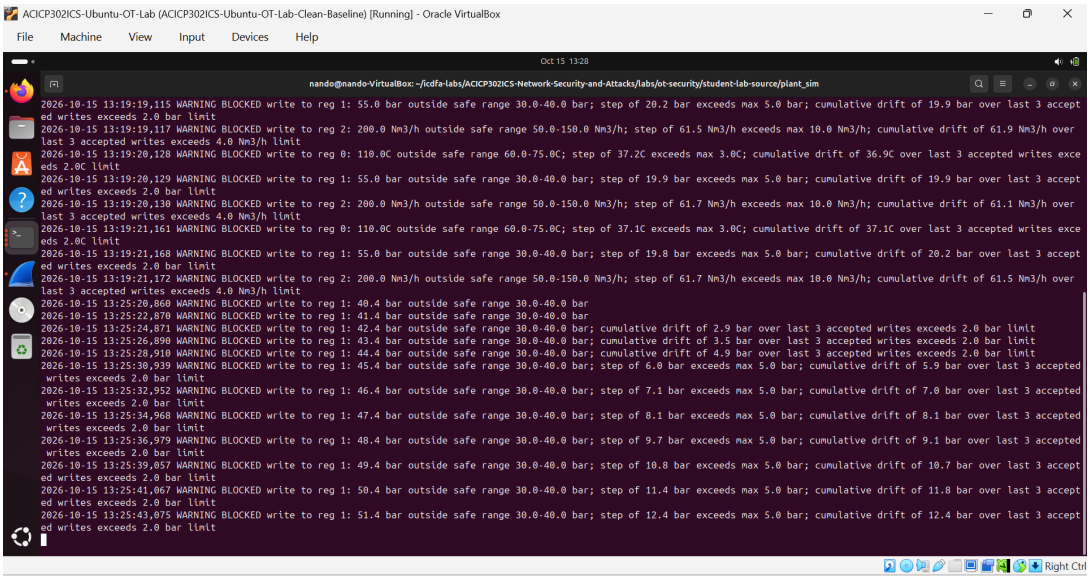


Figure 15: Slow-drift test – guard blocking evidence.

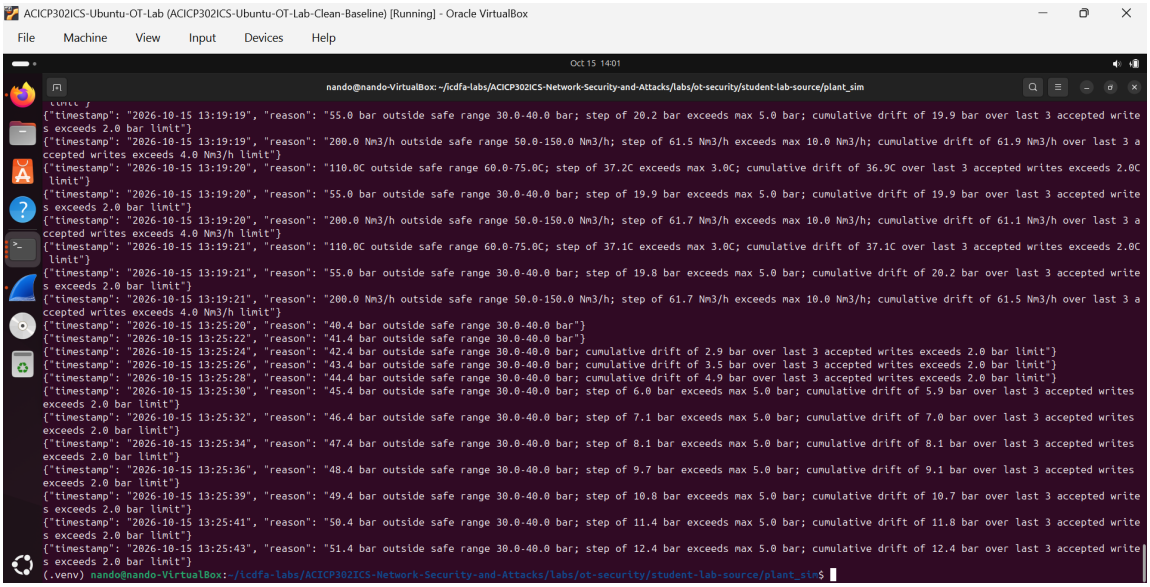


Figure 16: Guard event log showing false-data and slow-drift blocking events.

8.4 Test 3 – Replay Attack

The supplied replay script attempted to write the same pressure value three times. Observed output:

```
Replayed write 37.4 bar three times.
```

on the direct path, and:

```
Replayed write 37.7 bar three times.
```

on the guarded path.

No corresponding replay-related `BLOCKED write` event was observed in the supplied guard log.

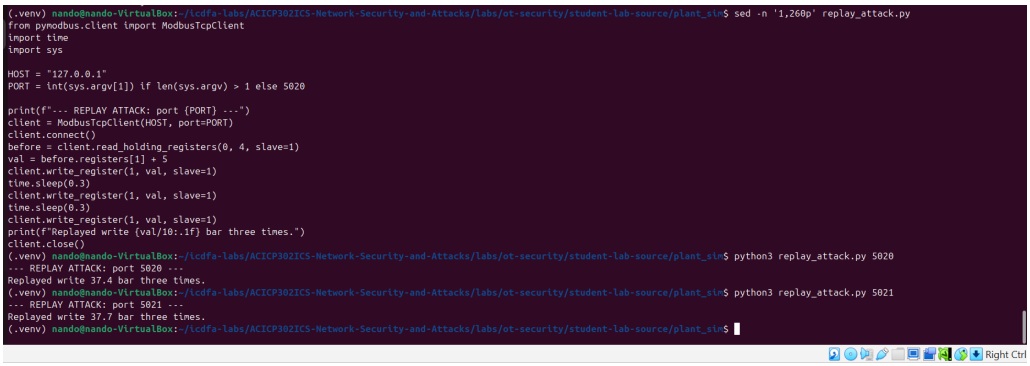


Figure 17: Replay demonstration result.

Result: The replay was not demonstrated as blocked by the supplied guard.

Security interpretation: This is a guard limitation rather than a failure of the test. The observed rule set evaluates process values, step size and drift, but does not provide explicit replay/duplicate transaction detection.

9 Test Comparison

Scenario	Direct path	Guarded path	Conclusion
False data	Unsafe values reached plant	Unsafe values blocked	Guard effective against demonstrated unsafe values
Slow drift	Pressure increased toward unsafe range	Unsafe writes blocked	Range, step and drift rules effective
Replay	Repeated write attempted	No replay block observed	Replay detection limitation

The three tests demonstrate that the guard provides useful rule-based protection, but it is not a complete OT security control.

10 Activity 07 – Recovery and Cleanup

After the demonstrations, the dashboards returned to normal operating ranges.

Dashboard	Temperature	Pressure	Flow
Guarded	72.6°C	39.4 bar	138.7 Nm^3/h
Direct	73.8°C	39.0 bar	132.5 Nm^3/h

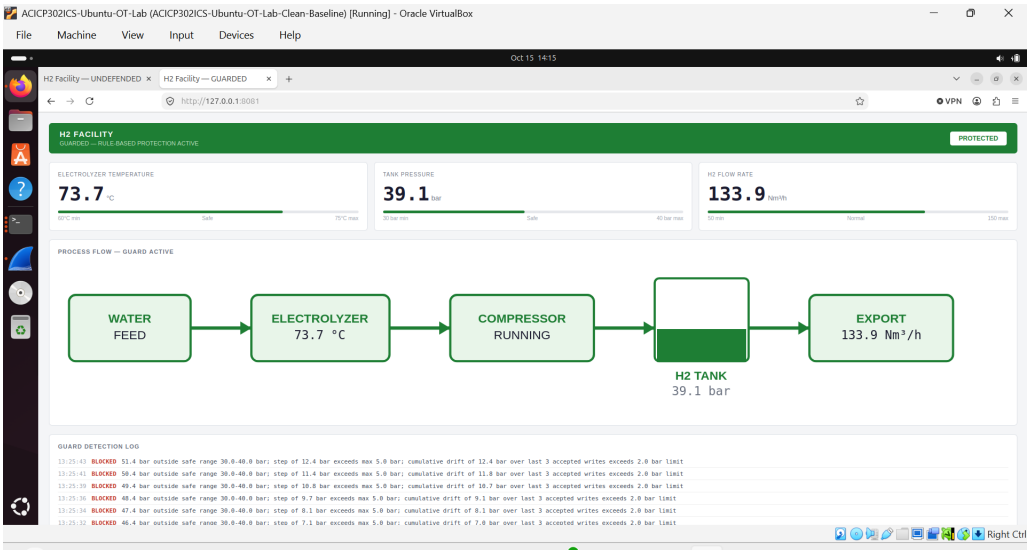


Figure 18: Recovered guarded dashboard.

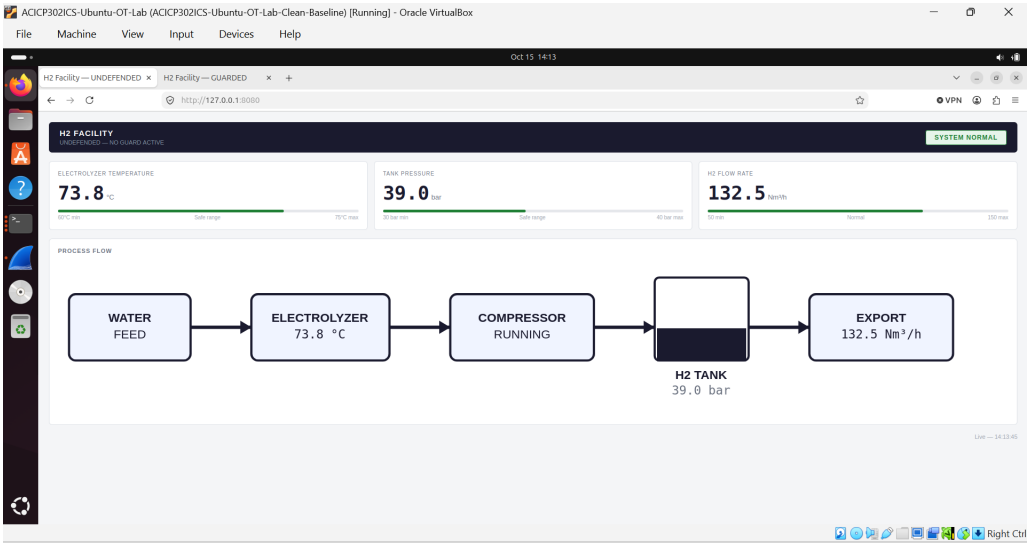


Figure 19: Recovered direct dashboard.

All four services were then stopped.

The expected lab ports were subsequently checked using:

```
ss -lntp | grep -E ':(5020|5021|8080|8081)\b'
```

No output was returned, confirming that the lab services were no longer listening.

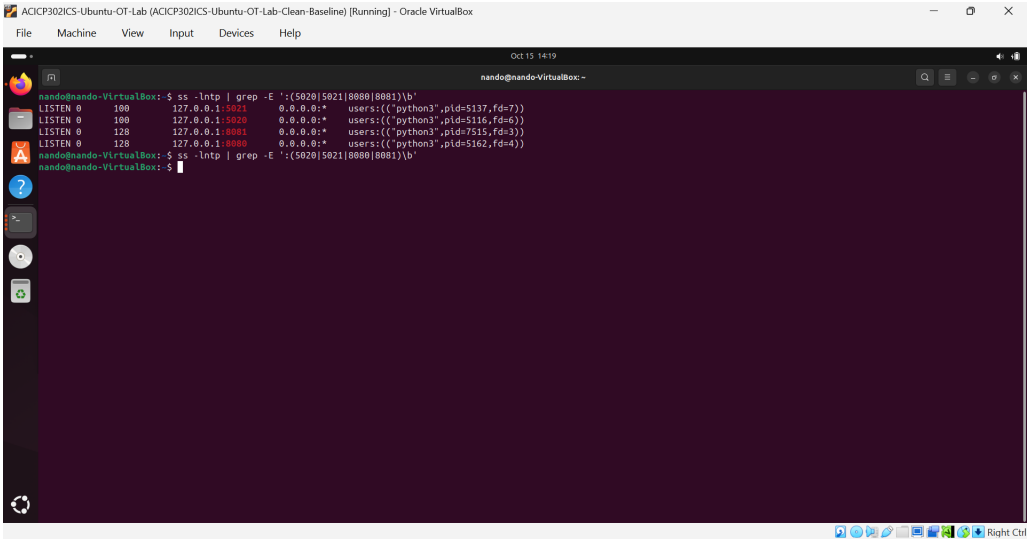


Figure 20: Verification that the lab ports were no longer listening.

The final network check confirmed that the VM remained isolated.

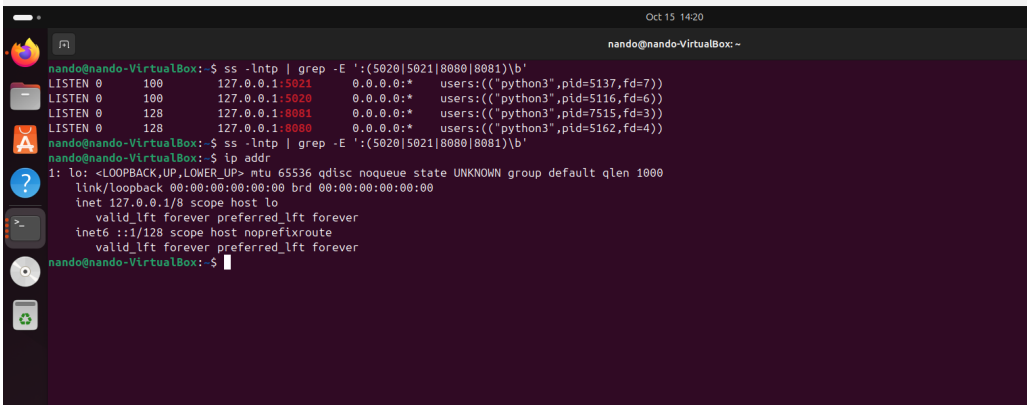


Figure 21: Final loopback-only network state.

11 Risk Register

ID	Risk	Evidence	Control / Mitigation	Level
R1	Unauthorized Modbus writes	Direct path accepted unsafe values	Restrict network access; use authenticated/authorized control paths	High
R2	Unsafe process-value manipulation	False-data demonstration	Range and step validation; monitor write events	High
R3	Gradual process manipulation	Slow-drift demonstration	Drift detection and threshold monitoring	High

R4	Replay/duplicate commands	Replay test produced no guard block	Add transaction/session/replay detection	Medium
R5	Threshold misconfiguration	Fixed guard limits	Validate thresholds against process requirements	Medium
R6	Availability impact	Direct Modbus service exposed locally	Network segmentation and controlled access	Medium

12 Activity 08 – Full Supplied-Source Run

The full hands-on run used the supplied student source under:

`labs/ot-security/student-lab-source/`

The source was executed using the project-local Python environment. The simulation was started, baseline operation was observed, normal Modbus traffic was captured, the supplied guard was analysed, three demonstrations were performed, and cleanup was verified.

No source modifications or threshold changes were made for the demonstrations.

The exact source commit recorded for the practical was:

`2a66f9bcf253eb3fe31c68ad52794cd535421da8`

13 Defence Summary

The practical can be defended using the following five points:

1. **Scope:** Testing was restricted to the assigned Ubuntu VM and loopback address `127.0.0.1`.
2. **Architecture:** The plant exposes Modbus/TCP on port 5020; the guard provides an intermediate path on port 5021.
3. **Guard effectiveness:** False-data and slow-drift attacks were blocked by range, step and cumulative-drift checks.
4. **Guard limitation:** The replay test produced no observed replay-specific block, demonstrating that the supplied guard does not provide explicit replay detection.
5. **Recovery:** Process values returned to normal ranges, all lab services were stopped, ports 5020, 5021, 8080 and 8081 were confirmed closed, and the VM remained isolated.

14 Conclusion

The practical demonstrated the security difference between a direct Modbus/TCP path and the supplied rule-based guarded path.

The guard successfully prevented the demonstrated unsafe false-data and slow-drift writes. However, the replay test showed that value-based controls alone do not provide complete protection against repeated valid-looking commands.

The exercise therefore demonstrates both the value and limitations of rule-based OT process protection. Recovery and cleanup were successfully verified after testing.